Q-INTEL

MODEL COMPARISON, VISUAL EVIDENCE & FINAL SELECTION

Candidate models, evaluation metrics, selection logic and charts for ETA regression and late-order classification.

Purpose
This document explains the candidate models, evaluation metrics, selection logic and charts for ETA regression and late-order classification. It is designed as the model-comparison deliverable for the hackathon.

Important version note
The numeric tables below reproduce the last fully recorded executed pipeline results available in the project history: a chronological split and a provisional **35-minute** late threshold. If the final notebook you are submitting has since changed to a 30-minute SLA or different candidate models, rerun that notebook and replace these figures before submission. **Do not mix metrics from different SLA/model versions.**

1. Evaluation design

The data was divided **chronologically** into training, validation and test periods. Candidate models were compared on validation data. The selected model was refitted using training plus validation data and evaluated once on the untouched test period. This reduces the risk of choosing a model based on future information.

Split	Rows	Share	Date range
Train	30,739	67.42%	11 Feb – 23 Mar 2022
Validation	7,525	16.50%	24 Mar – 30 Mar 2022
Test	7,329	16.08%	31 Mar – 06 Apr 2022

Why these metrics?

Regression
MAE is average absolute error in minutes and is easiest to explain; **RMSE** penalises large ETA misses; R^2 indicates variance explained relative to a mean baseline. Lower MAE/RMSE and higher R^2 are preferred.

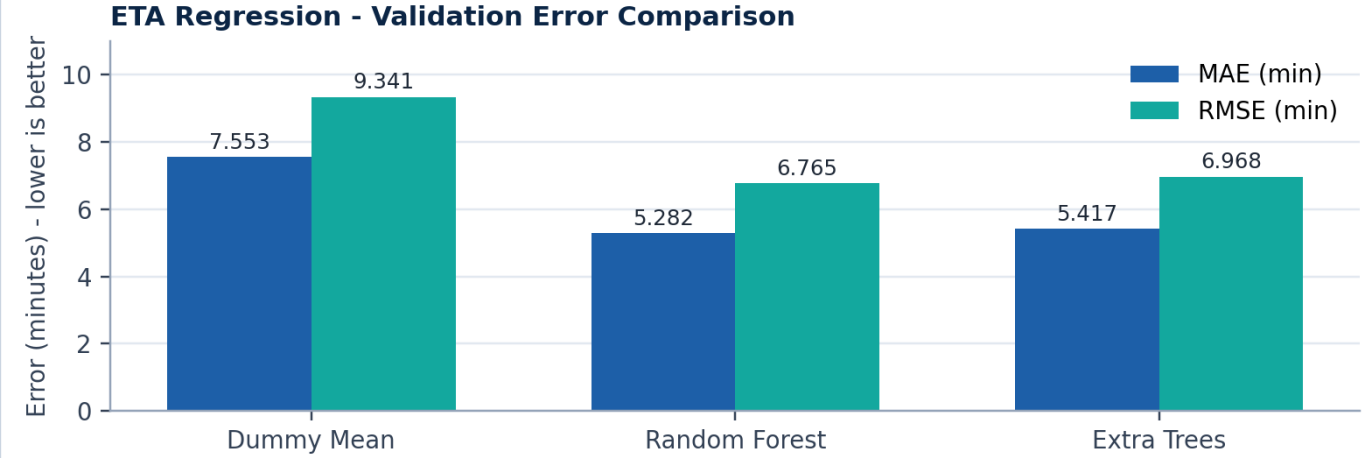
Classification
Precision measures how many flagged orders were actually late; **recall** how many actual late orders were caught; **F1** balances the two; **PR-AUC** summarises precision-recall across thresholds. **ROC-AUC** shows ranking quality. Accuracy alone can hide missed late orders.

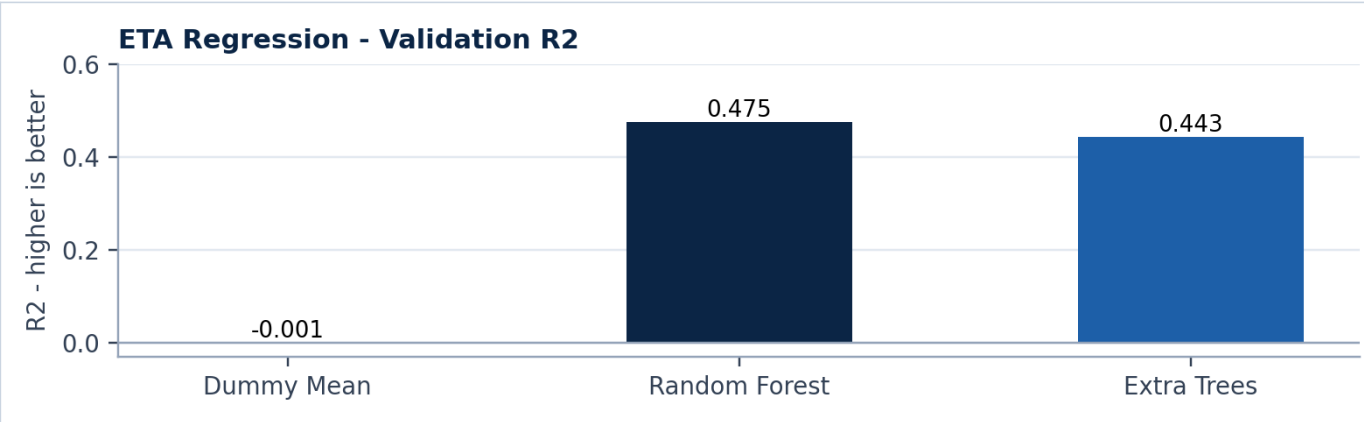
Decision discipline
Choose models/thresholds on **validation** evidence, then report test results **once**. Never tune on the test set.

2. ETA Regression — Model Comparison

Target: delivery duration in minutes. Candidate models in the recorded pipeline were Dummy Mean, Random Forest and Extra Trees. The Dummy model predicts the training-set mean and acts as a baseline.

Model	Validation MAE (lower)	Validation RMSE (lower)	Validation R^2 (higher)	Interpretation
Dummy Mean	7.553	9.341	-0.001	Reference baseline; almost no explanatory value.
Random Forest	5.282	6.765	0.475	Lowest error and highest R^2 among the compared candidates.
Extra Trees	5.417	6.968	0.443	Better than baseline, but slightly weaker than Random Forest.





Selection: Random Forest Regressor
It achieved the lowest validation MAE (5.282 minutes), lowest RMSE (6.765) and highest R^2 (0.475) among the candidates. This is a metric-led selection, not a claim that Random Forest is universally best.

Why not Gradient Boosting?
It is a valid candidate to test, but it was not included in this recorded run's comparison table. Do not claim it was tested or rejected unless it appears in the final notebook output.

3. Selected ETA Model — Held-Out Test Result

After selecting Random Forest on validation data, the model was refitted on the combined training and validation data and evaluated on the chronological test period.

MAE (min)	RMSE (min)	R^2
5.2651	6.6921	0.495

Chronological held-out test set | Selected ETA model: Random Forest

Test metric	Recorded result	Plain-language meaning
MAE	5.2651 minutes	On average, the absolute difference between predicted and actual delivery time was about 5.27 minutes.
RMSE	6.6921 minutes	Larger prediction errors receive more penalty; useful for detecting occasional large misses.
R^2	0.4950	The model explained about 49.5% of the target variance on this test split.

Validation vs held-out test (Random Forest)

Dataset	MAE (min)	RMSE (min)	R^2
Validation	5.282	6.765	0.475
Test	5.2651	6.6921	0.4950

Interpretation for the judging panel

The Random Forest improves materially over the mean baseline. However, an MAE of approximately 5.27 minutes may still be significant in a quick-commerce setting. The model should be treated as decision support and evaluated against the actual service promise before operational use.

Feature importance context

The recorded permutation-importance analysis identified road-traffic density, weather and delivery distance among the strongest ETA predictors. Permutation importance measures predictive reliance, not causation; it does not prove that changing a factor will reduce delays.

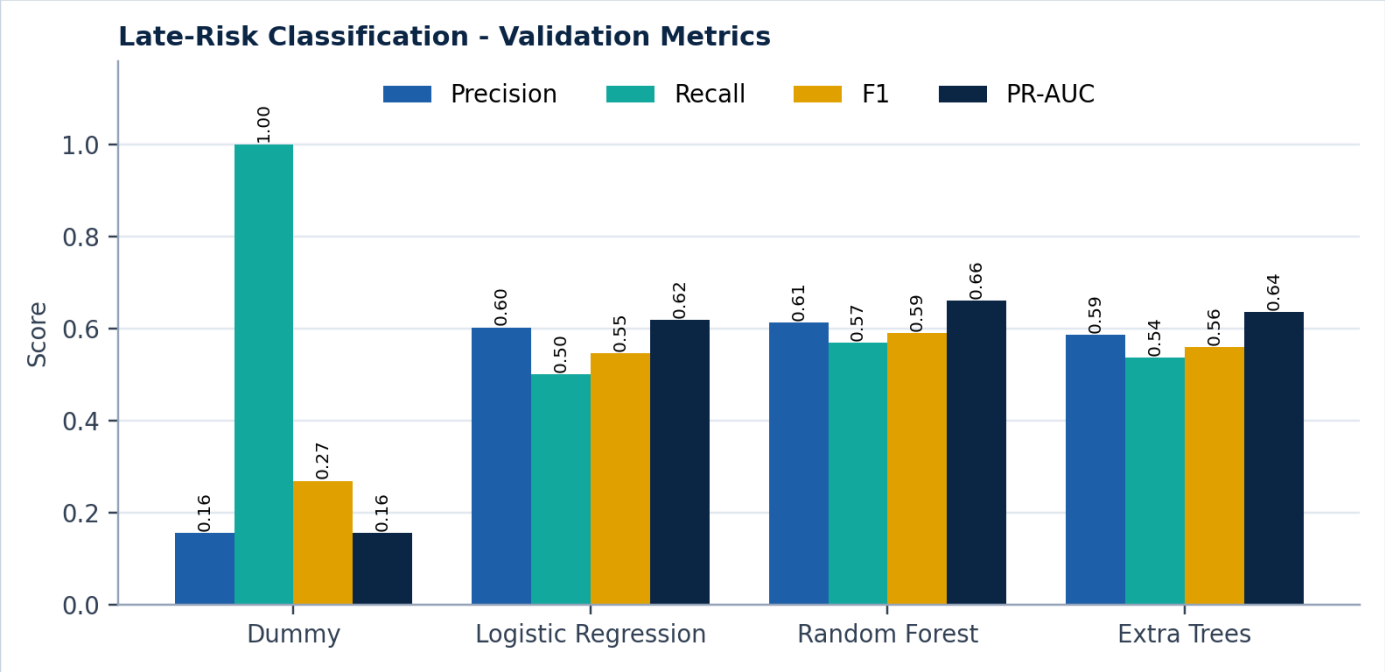
Limitations

The source is food-delivery data used as a quick-commerce proxy. Restaurant coordinates stand in for stores. Straight-line GPS distance is not road distance. The 35-minute threshold is provisional and must be replaced if the organiser approves a different SLA.

4. Late-Order Risk — Classifier Comparison

Target: 1 when actual delivery duration exceeds the provisional 35-minute threshold. Candidate models were Dummy, Logistic Regression, Random Forest and Extra Trees. Each non-dummy classifier's threshold was selected on validation data by maximising F1 in the recorded run.

Model	Threshold	Precision	Recall	F1	PR-AUC
Dummy	0.000	0.1553	1.0000	0.2689	0.1553
Logistic Regression	0.7449	0.6019	0.5004	0.5465	0.6183
Random Forest	0.5030	0.6123	0.5689	0.5898	0.6595
Extra Trees	0.5884	0.5854	0.5364	0.5598	0.6357



Selection: Random Forest Classifier
Random Forest had the best F1 (0.5898) and PR-AUC (0.6595) among the compared non-dummy models, with the highest precision (0.6123) and recall (0.5689) in this validation comparison. The threshold was approximately 0.503. The dummy model's recall of 1.0 is misleading: it flags every order as late and has very low precision.

Business trade-off
A lower threshold usually catches more late orders but creates more false alerts. A higher threshold reduces unnecessary interventions but may miss more late orders. The saved threshold is a validation-selected operating point, not a universal optimum. A 5:1 missed-late vs unnecessary-intervention cost ratio should only be stated if it is actually used in the final notebook's threshold calculation.

5. Selected Classifier — Test Result & Recommendation

Test metric	Recorded result	Meaning
Accuracy	0.8509	85.09% of all test predictions were correct; not sufficient on its own.
Precision	0.5870	About 58.7% of orders flagged late were actually late.
Recall	0.6277	About 62.8% of actual late orders were detected.
F1	0.6067	Balance between precision and recall.
ROC-AUC	0.8814	Good ranking separation between late and non-late examples.
PR-AUC	0.6804	Precision-recall performance across thresholds.

Confusion matrix

	Predicted on-time	Predicted late
Actual on-time	**5,393** true negatives	**593** false positives
Actual late	**500** false negatives	**843** true positives

What this means operationally
The model detected **843 of 1,343** actual late orders, while missing **500**. It also raised **593 false alerts**. Operations must decide whether this balance is acceptable given staffing capacity and the cost of a missed late order. Validate the threshold with a controlled pilot and monitor alert volume.

Final model decision

Task	Selected model	Reason
ETA regression	Random Forest Regressor	Best validation MAE/RMSE and R^2; test MAE 5.2651 minutes.
Late-risk classification	Random Forest Classifier	Best validation F1 and PR-AUC among candidate classifiers; test F1 0.6067 and ROC-AUC 0.8814.

Submission note — reconcile before final upload
This PDF accurately records the earlier fully documented 35-minute-SLA pipeline. Your project context also mentions a later revised 30-minute SLA and updated modelling workflow. The current final notebook's executed validation tables and test outputs were not available in the retrieved notebook text (its cells were unexecuted). If your final version differs, rerun it and update every metric/chart here before submitting. The method remains: compare candidates on validation, select using business-appropriate metrics, refit on train + validation, and evaluate once on the untouched test set.

Hackathon alignment
This comparison addresses the challenge requirement to show a baseline, compare at least two regression models, evaluate late-risk classification using precision/recall/F1/ROC-AUC, and justify final selections. It should be accompanied by K-Means elbow/silhouette evidence and the zone/rider analysis from the final notebook.